

MBA

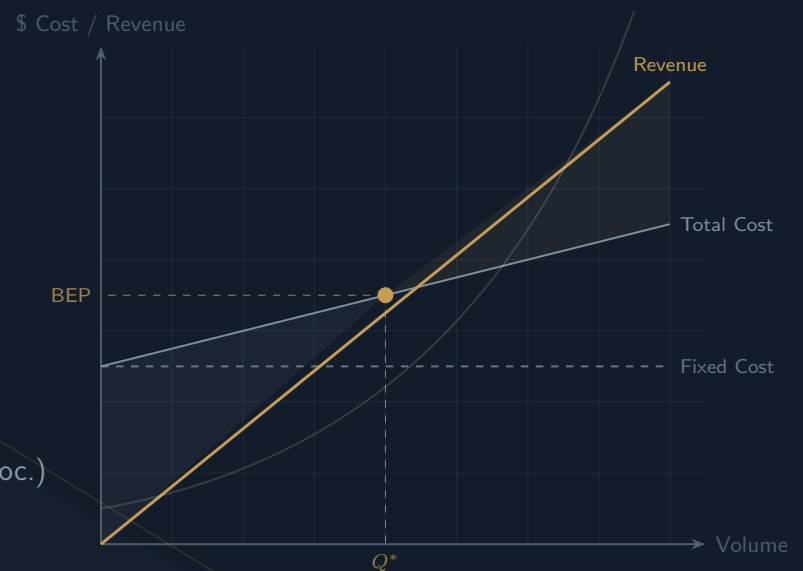
MATHEMATICS

CLINIC

$NPV = \sum_{t=1}^T \frac{CF_t}{(1+r)^t} - C_0$   
Quantitative Skills for Business Decision-Making

$$\% \Delta = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100$$

Profit = Revenue - Cost



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# MBA Mathematics Clinic

## Quantitative Skills for Business

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**MBA Mathematics Clinic: Quantitative Skills for Business**

This clinic is designed as an active problem-solving session rather than a conventional lecture. Attempt each question before reading the solution.

## 1 How the Clinic Will Run

The aim is not to cover a large amount of theory. The aim is to practise the numerical and algebraic skills that repeatedly appear in accounting, finance, economics, and managerial decision-making.

**Principle 1.1 (Attempt first, then discuss)**

For each exercise, first identify the information given, decide what quantity is being asked for, write the mathematical relationship, and only then perform the arithmetic. A numerical answer without an interpretation is incomplete in a business setting.

Our two-hour work flow are as follows:

1. a short diagnostic warm-up;
2. percentages, ratios, and weighted quantities;
3. algebra and formula manipulation;
4. structured accounting and business questions;
5. interpretation and discussion of the answers.

## 2 Part I: Numerical Fluency and Percentages

### 2.1 A Short Diagnostic

**Exercise 2.1 Numerical warm-up**

Evaluate each of the following without skipping steps.

- (a)  $48 - 3(7) + 16 \div 4$ .
- (b)  $\frac{3}{4} + \frac{5}{6}$ .
- (c)  $\left(\frac{3}{4} + \frac{5}{6}\right) \div \frac{7}{3}$ .
- (d) Find 18% of \$1,250.
- (e) A price of \$1,250 is reduced by 18%. Find the new price.

## 2.2 Percentage Change

### Exercise 2.2 Successive percentage changes

Monthly sales increase from \$240,000 to \$276,000. In the following month, sales fall by 10% from the new level.

- (a) Find the percentage increase in the first month.
- (b) Find the sales figure after the 10% fall.
- (c) Relative to the original \$240,000, find the overall percentage change.
- (d) Explain why a 15% increase followed by a 10% decrease does not give a net increase of exactly 5%.

### Exercise 2.3 Reverse percentage

After a 15% discount, a piece of equipment is sold for \$1,020.

- (a) Find the original listed price.
- (b) A student claims that the original price is  $1020 + 15\%(1020)$ . Explain the error in this reasoning.

## 3 Part II: Ratios, Proportions, and Weighted Quantities

### 3.1 Ratio Allocation

#### Exercise 3.1 Allocating a budget by ratio

A department has a budget of \$840,000 to be divided among Operations, Marketing, and Technology in the ratio

$$3 : 4 : 5.$$

- (a) Determine the allocation to each department.
- (b) What percentage of the total budget is allocated to Technology?

### 3.2 Weighted Averages

#### Exercise 3.2 Weighted profit margin

A company earns revenue from three divisions. Division A produces 40% of total revenue and has a profit margin of 25%. Division B produces 35% of revenue and has a profit margin of 18%. Division C produces the remaining revenue and has a profit margin of 12%.

- (a) Find the proportion of revenue produced by Division C.
- (b) Find the company-wide weighted profit margin.
- (c) Explain why simply averaging 25%, 18%, and 12% would generally be inappropriate.

## 4 Part III: Algebra for Business

### 4.1 Solving Equations

#### Exercise 4.1 Solve for the unknown

Solve

$$0.6x + 120 = 0.8x - 40.$$

Then verify your answer by substituting it into both sides of the equation.

### 4.2 Rearranging Business Formulae

#### Exercise 4.2 Profit and target sales volume

Suppose

$$R = pq, \quad C = F + vq, \quad P = R - C,$$

where  $p$  is selling price per unit,  $q$  is quantity sold,  $v$  is variable cost per unit,  $F$  is fixed cost, and  $P$  is profit.

(a) Show that

$$P = (p - v)q - F.$$

(b) Rearrange the formula to make  $q$  the subject.

(c) State the condition required for the rearranged formula to make economic sense.

## 5 Part V: Time Value of Money

### 5.1 Compound Growth and Present Value

#### **Exercise 5.1** Compound interest

An investment of \$500,000 earns 8% per year compounded annually for three years.

- (a) Write the compound-interest model.
- (b) Find the value after three years.
- (c) Find the total interest earned over the three years.

#### **Exercise 5.2** Present value

A business expects to receive \$750,000 four years from now. If the appropriate annual discount rate is 9%, find the present value of that future amount.



## 6 Part VI: Integrated Decision Making

### Exercise 6.1 Should the company offer the discount?

A company currently sells a service package for \$250. Variable cost is \$140 per package, monthly fixed cost is \$330,000, and expected monthly sales are 4,200 packages. Management is considering an 8% price reduction. It estimates that the discount would increase sales volume by 12%.

- (a) Find current monthly profit.
- (b) Find the proposed discounted selling price.
- (c) Find the expected sales volume after the projected 12% increase.
- (d) Find expected monthly profit under the discount proposal.
- (e) By how much would monthly profit change?
- (f) At the discounted price, determine the minimum whole number of packages that must be sold to match the current monthly profit.
- (g) What percentage increase over the original 4,200 packages would this require?
- (h) Based only on this cost-volume information, should management adopt the proposal? Give a quantitative justification.

## 7 Part VII: Discussion and Interpretation

The following questions are intended to be answered verbally and defended with reasoning.

### Exercise 7.1 Discussion questions

For each statement, decide whether you agree or disagree and justify your answer.

- (a) "Revenue increased, so the business must be performing better."
- (b) "A 20% decrease followed by a 20% increase returns a quantity to its original value."
- (c) "A product with a 30% markup has a 30% profit margin."
- (d) "If actual expenses are below budget, management performance must be favourable."
- (e) "A current ratio above 1 automatically means that the firm has no liquidity problem."

## 8 Mathematics in the Trinidad and Tobago Business Environment

In practice, managers routinely use percentages, ratios, linear models, financial measures, and rates of change to interpret the performance of firms and to support business decisions.

In this section, we apply some of these ideas to publicly reported data from major organisations operating in Trinidad and Tobago. The objective is not to perform a detailed financial analysis of these companies, but rather to use realistic business data to develop quantitative reasoning.

### 8.1 Case Study 1: Massy Holdings Ltd.

Massy Holdings Ltd. is a Trinidad and Tobago-founded conglomerate with operations across several sectors, including integrated retail, motors and machines, and gas products.

Selected financial information is shown below.

| Measure                             | 2023   | 2024   | 2025   |
|-------------------------------------|--------|--------|--------|
| Revenue (TT\$ billions)             | 13.801 | 15.354 | 15.833 |
| Profit before tax (TT\$ billions)   | 1.206  | 1.096  | 1.141  |
| Debt-to-equity ratio                | 46%    | 37%    | 34%    |
| Operating cash flow (TT\$ billions) | 1.018  | 1.349  | 1.670  |

#### Exercise 8.1 Revenue Growth and Profitability

Using the data above:

1. Calculate the absolute increase in revenue from 2024 to 2025.
2. Calculate the percentage increase in revenue from 2024 to 2025.
3. Calculate the profit-before-tax margin for:
  - (a) 2024;
  - (b) 2025.
4. Based on your answers, did profitability improve at approximately the same rate as revenue?
5. The debt-to-equity ratio decreased from 46% in 2023 to 34% in 2025.
  - (a) Calculate the decrease in *percentage points*.
  - (b) Calculate the percentage reduction relative to the 2023 value.
  - (c) Explain why these two answers are not the same.

## 8.2 Case Study 2: The National Gas Company of Trinidad and Tobago

The National Gas Company of Trinidad and Tobago Limited (NGC) operates within the country's energy sector.

Suppose the following publicly reported figures are considered:

- profit after tax in 2024: TT\$1.6 billion;
- loss after tax in 2023: TT\$1.3 billion;
- outstanding TT&TEC receivables: approximately TT\$6.3 billion;
- estimated annual increase in those receivables if the situation continues unchanged: approximately TT\$1.2 billion per year.

### Exercise 8.2 From Loss to Profit

NGC moved from a loss of TT\$1.3 billion in 2023 to a profit of TT\$1.6 billion in 2024.

1. Represent the 2023 loss as a negative number and calculate the absolute improvement in financial performance between 2023 and 2024.
2. A manager attempts to describe the movement from the 2023 loss to the 2024 profit using an ordinary percentage-change formula. Explain why this calculation should be interpreted with caution.
3. Explain, in business terms, why moving from a loss to a profit may be more appropriately described using the absolute turnaround rather than an ordinary percentage increase.

### Exercise 8.3 Building a Linear Model

Suppose outstanding receivables are currently TT\$6.3 billion and are expected to increase by TT\$1.2 billion per year if the current situation remains unchanged.

Let  $R(t)$  denote the outstanding receivables, in billions of Trinidad and Tobago dollars, after  $t$  years.

1. Construct a linear model for  $R(t)$ .
2. Estimate the outstanding receivables after:
  - (a) one year;
  - (b) three years;
  - (c) five years.
3. According to the model, after how many years would the outstanding amount exceed TT\$10 billion?
4. State one limitation of this model.

### 8.3 Case Study 3: Methanex Trinidad

Operational data can also be used to demonstrate the importance of percentages and ratios.

Suppose Methanex's Trinidad operations had an operating capacity, corresponding to its interest, of approximately 1.960 million tonnes.

Production was approximately

956 000 tonnes

in 2024, compared with

1 074 000 tonnes

in 2023.

#### Exercise 8.4

#### Production and Capacity Utilisation

Using the information above:

1. Calculate the decrease in production from 2023 to 2024.
2. Calculate the percentage decrease in production.
3. Calculate the 2024 capacity-utilisation rate using

$$\text{Capacity Utilisation} = \frac{\text{Actual Production}}{\text{Operating Capacity}} \times 100\%.$$

4. Suppose management wished to operate at 70% of capacity. Calculate:
  - (a) the corresponding target production;
  - (b) the additional production required relative to the 2024 level.
5. Briefly explain why capacity utilisation may be an important measure for a manager in an industrial or energy-sector organisation.

## 8.4 Discussion: What Does the Number Actually Mean?

The calculations in the previous case studies illustrate an important distinction between *calculation* and *interpretation*.

A manager must be able not only to compute a numerical answer, but also to decide whether the number is meaningful and what conclusions can reasonably be drawn from it.

### Exercise 8.5 Management Discussion

Consider the three case studies above.

Discuss the following questions.

1. Why can revenue increase while a firm's profit margin remains almost unchanged?
2. Why is a decrease of 12 percentage points different from a 12% decrease?
3. Why can an ordinary percentage-change calculation become misleading when the original value is negative?
4. What assumptions are being made when a linear model such as

$$R(t) = 6.3 + 1.2t$$

is used to project future receivables?

5. Does a low capacity-utilisation percentage necessarily imply poor management? Explain.
6. Which is more important for a manager: obtaining the correct numerical answer, or understanding the assumptions and business meaning of the calculation?